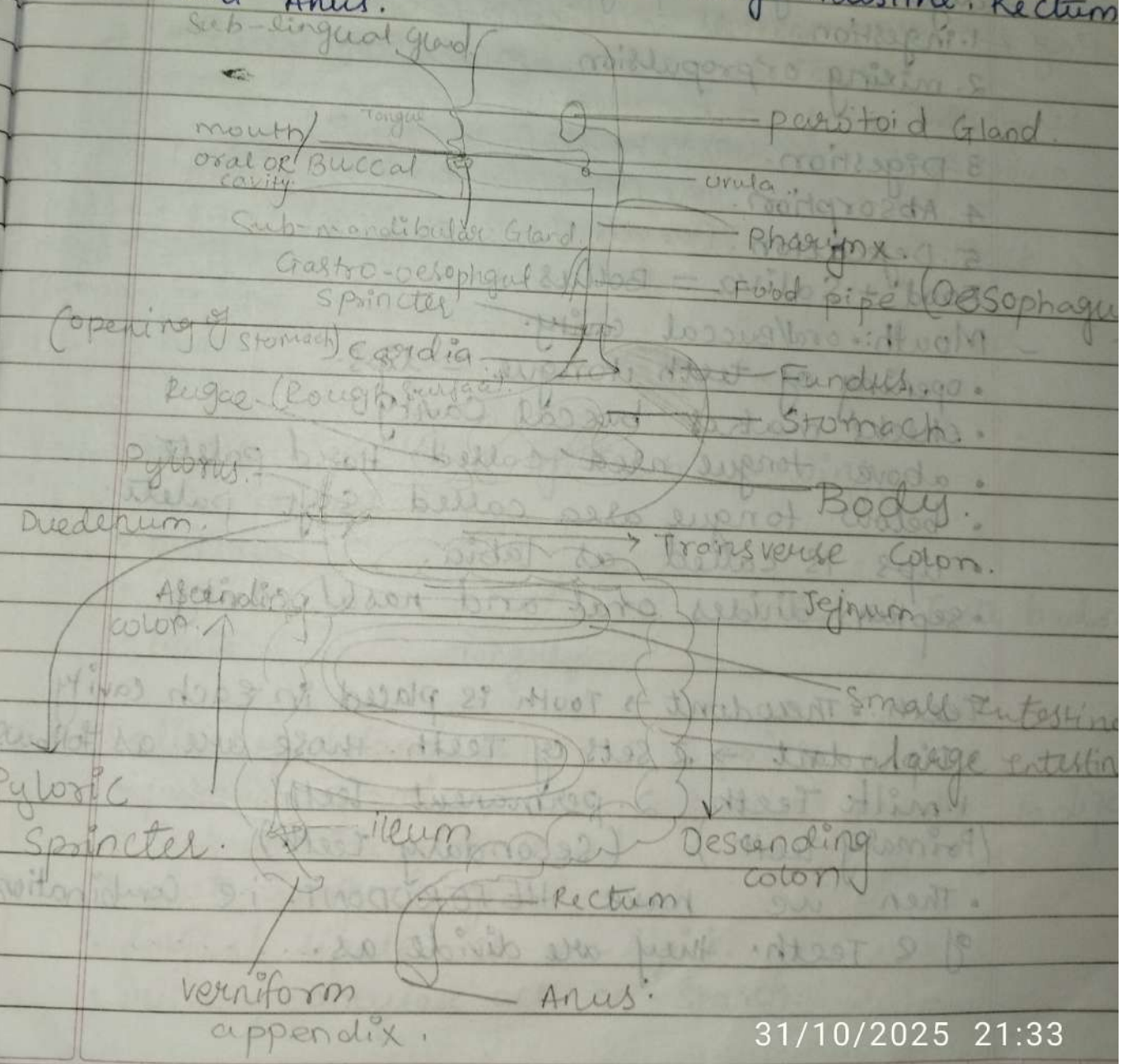


muscles → cardiac → Heart.
 → Smooth → Can't move.
 → Skeleton → movable.

* DIGESTIVE SYSTEM:

- Elementary canal → mouth to Anus.
- 30% of food gets digested in mouth only.
- Digestion → complex molecule to simple molecule.
- Food absorbs in the form of glucose in the blood.

- A human digestive system consists of: mouth, Teeth, Salivary glands, pharynx, Esophagus, Liver, Stomach, pancreas, small intestine, large intestine, Rectum And Anus.



- Digestive System is majorly divided into GI i.e. Gastrointestinal Tract and Accessory digestive organ.

- G.T. has mouth, pharynx, Esophagus, Stomach, Small intestine, large intestine.

- Accessory Digestive organs are Teeth, Tongue, Salivary glands, liver, Gall bladder & pancreas.

- Digestive System performs Five basic process.

- Eating.
1. ingestion. (Food and liquids taking into the mouth)
 2. mixing. or propulsion. (generation of force by any combination of pushing).
 3. Digestion. (Break down of food).
 4. Absorption. (The process of taking something into another substance)
 5. Defecation. (The discharge of feces from the body)

- Food + Saliva = Bolus.

- Mouth: • oral/Buccal cavity.

- opening of teeth & tongue = lips.
- cheeks makes buccal cavity.
- above tongue area called Hard palate.
- below tongue area called Soft palate.
- lips is called as labia.
- septum divides oral and nasal cavity.

- Teeth: • Therodont → Tooth is placed in Each cavity.

• Dipodont → 2 sets of Teeth. those are as follows.

1. milk Teeth. (Primary Teeth)
2. permanent Teeth. (Secondary Teeth)

• Then we have HETERODONT. i.e. Combination of 2 Teeth. they are divide as.....

It is the only one side from 4

HETERODONT

Incisors (I)
(Rabbit Teeth)
[2]

Canines (C)
(Pointed Teeth)
[1]

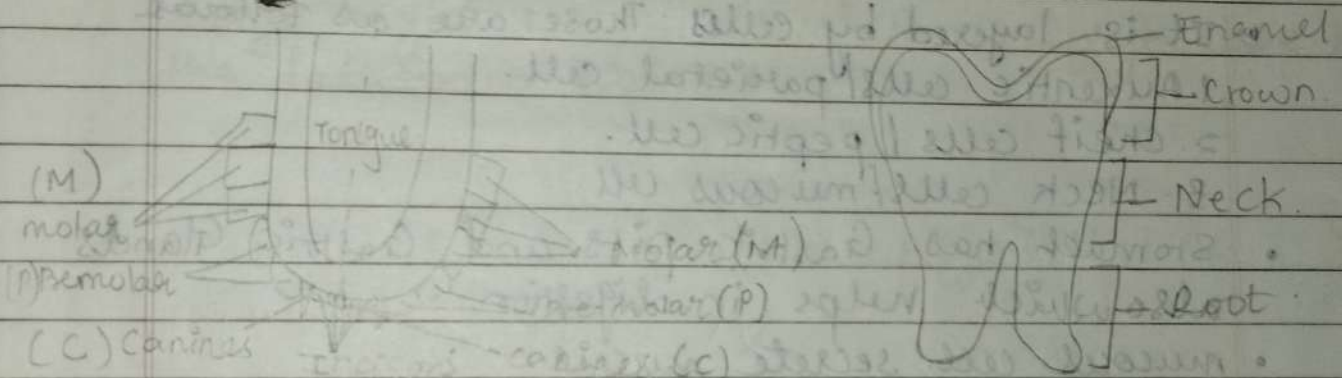
Premolar (P)
(used in Eating)
[2]

Molar (M)
(wisdom Tooth)
[3]

Dental Formula: $\frac{2123}{2123}$ — upper Jaw
ICPM — lower Jaw

• upper Jaw — maxilla — 16 Teeth.

• lower Jaw — mandible — 16 Teeth.



Tongue: • Functions of Teeth: • chewing.

• Swallowing.

• Bolloos.

• Speech

• Taste

Tongue //

Papillae.

called as Taste buds.

— Salivary Glands: • It has 3 pairs.

1. parotid gland.

2. Sub-mandibular gland.

3. Sub-lingual gland.

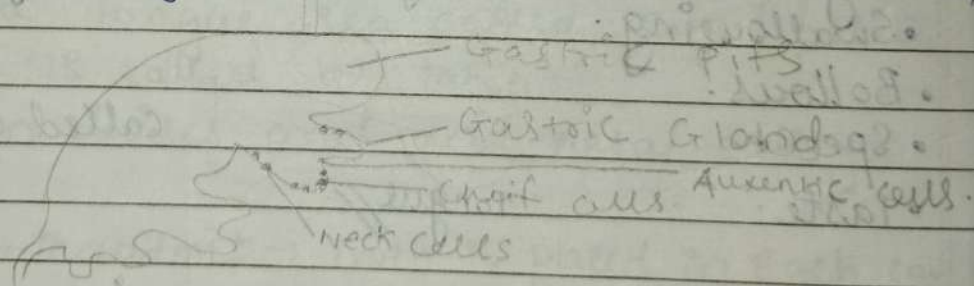
• lingual lipase acts on - lipid.

• Salivary amylase acts on - starch.

- Oesophagus: • It is 25cm. long in length.
- also called as food pipe.
- peristaltic movement: contraction in oesophagus when food passes.
- Secretes mucus.

→ Stomach: • It has 4 main regions: 1. Cardia (opening of Stomach)
2. Fundus (rounded portion left to Cardia)
3. Body (large central portion of Stomach)
4. Pylorus (region that connects to the duodenum)

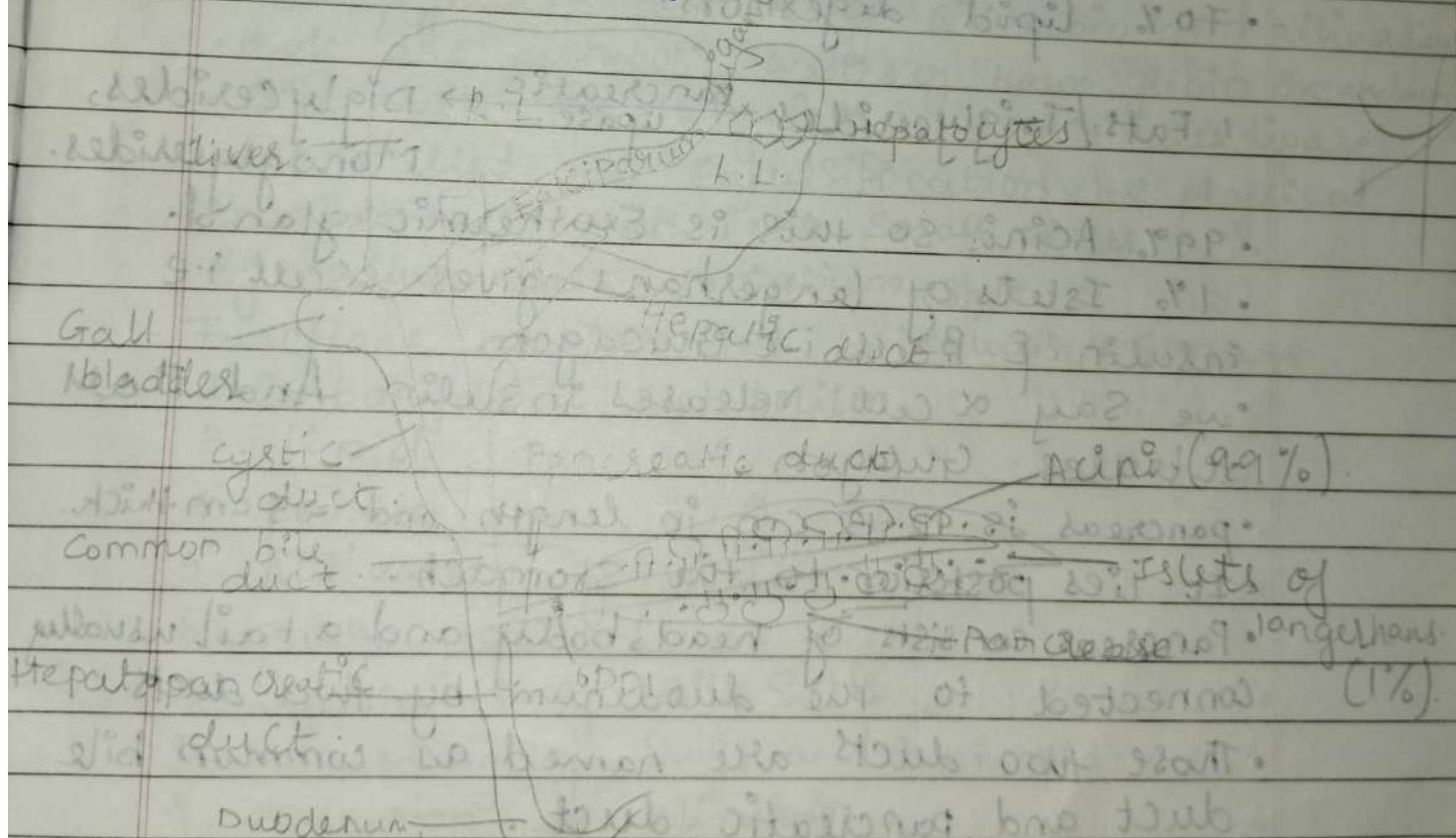
- It is layered by cells: Those are as follows.
 1. Acidic cells / parietal cell.
 2. Chief cells / peptic cell.
 3. Neck cells / mucous cell.
- Stomach has Gastric pits and Gastric Glands also which helps in digestion.
- mucous cell secrete mucus.
- Chief cell secrete pepsinogen and gastric lipase i.e gastric juice which totals 2000-3000ml/day.



- The Glands which releases through duct is Exocrine gland.
- The Glands which releases without duct is Endocrine gland.
- Swallowing of the food is deglutition.
- Synocobalamin is vit B12.

- parietal cell produces intrinsic factor and hydrochloric acid (HCl.)
- muscularis masticate food and mix it with gastric juice, forming chyme.

- liver: • It is the largest gland of the human body.
- Its personal weight is 1.2-1.5 kg.
- liver converts nitrogenous waste ammonia (NH_3) into urine.
- It is inferior to the diaphragm.
- liver is divided into two principle lobes: 1 is a large right lobe and a smaller left lobe by the falciform ligament.



HEPATOPANCREATIC SYSTEM.

- Right lobe and left lobe are connected with Falciform ligament.

Stomach → Gastric
lungs → pulmonary.
Liver → Hepatic
Gall bladder → cystic.

— pancreas: • pancreas releases pancreatic juice.

• Its pH is 8.6 i.e. Alkaline. (alkaline)

• Protein digestion takes place.

1. Trypsinogen $\xrightarrow{\text{Entero kinase}}$ Trypsin

2. Chymotrypsinogen $\xrightarrow{\text{Trypsin}}$ Chymotrypsin

3. Procarboxypeptidase $\xrightarrow{\text{Trypsin}}$ Carboxypeptidase

• Carbohydrate digestion.

1. Polysaccharides $\xrightarrow{\text{Pancreatic amylase}}$ Disaccharides.

• 70% lipid digestion:

1. Fats / Triglycerides $\xrightarrow{\text{Pancreatic lipase}}$ Diglycerides, Monoglycerides.

• 99% Acini; so this is Exocrine gland.

• 1% Islets of Langerhans gives α cell i.e. insulin & β cell i.e. Glucagon.

• we say α cell releases insulin And β cell releases Glucagon.

• pancreas is 12-15 cm in length and 2.5 cm thick.

• It lies posterior to the stomach.

• Pancreas consists of head, body and a tail usually connected to the duodenum by two ducts.

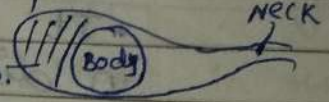
• Those two ducts are named as common bile duct and pancreatic duct.

• where these two ducts meet is called as Hepatopancreatic duct.

• pancreas is the one and only heterocrine gland i.e. mixture of Endocrine and Exocrine glands.

PH $\rightarrow$ Potential of Hydrogen.

- Gall bladder: • It is a pear shaped.
 - It is located in a depression of the posterior surface of the liver.
 - It is 7-10 cm long in length.
 - Typically hangs from the anterior inferior margin of the liver.
 - Gall bladder includes the broad fundus.
 - The duct come from Gall bladder to common bile duct is cystic duct.



Gall bladder stores & concentrate the bile juice until needed in small intestine.

- Bile: • Each day hepatocytes secrete 1000-1500 ml of bile.
 - Bile has majorly Bile pigments, Bile Salts, Cholesterol and phospholipids.
 - Bile pigments consists of Bilirubin and Biliverdin.
 - These are metabolic wastes of Haemoglobin Breakdown.
 - Bile salts Emulsify fats and Activates lipase. also called as Emulsification. i.e physical Breakdown of fats into small pieces.
 - Eg: Sodium Salt, Potassium Salt.

- Functions of liver are as follows:
 1. Storage.
 2. Carbohydrate metabolism.
 3. Lipid
 4. Protein
 5. Activation of vit D.
 6. Phagocytosis. (?)
 7. Synthesis of Bile Salts.
 8. Processing of drugs and hormones.
 9. Excretion of bilirubin.

- Small Intestine : • Major digestion Takes place here only.
- It is 3 cm in length and 2.5 cm average in diameter in a living person.
- Small intestine divides into 3 regions. Those are as follows: 1. Duodenum. 2. Jejunum 3. ileum.
- from which duodenum is the shortest region in the start.
- Jejunum is 1 m long and extends to the ileum.
- The final and the longest region of small intestine, is the ileum.
- It is 2 m long in length and joins to large intestine.
- mucosa and submucosa intestinal glands secrete intestinal juice.
- Duodenal i.e. Brunner's gland secretes alkaline fluid to buffer stomach acids and mucus for protection and lubrication.
- microvilli are the microscopic membrane-covered projections.
- villi are the finger-like projections of mucosa.
- Circular folds of mucosa and submucosa increase the surface area for digestion and absorption.
- The absorptive cells of the small intestine synthesize several digestive enzymes called brush border enzymes. And these are as follows: α -Dextrinase, Maltase, Sucrase, Lactase, Enterokinase, peptidases, Amino peptidase, Dipeptidase, nucleosidases and phosphatases.

- The type of peristaltic movement that occurs in small intestine is termed as migrating motility complex (MMC).
- Any undigested or unabsorbed material left in the small intestine passes on to the large intestine.
- crypts of Lieber Kuhn intestinal glands releasing.
- Succus Entericus present on the last layer of intestine.
- PH of Succus Entericus is 7.8
- Secretions of Small intestine:
 - Goblet cells - Secrete mucus.
 - Carbohydrates - Converts Everything to monosaccharide.
 1. Maltose maltase → Glucose + Glucose.
 2. Lactose lactase → Glucose + Galactose.
 3. Sucrose sucrase → Glucose + Fructose.
 - Nucleic acid digestion:

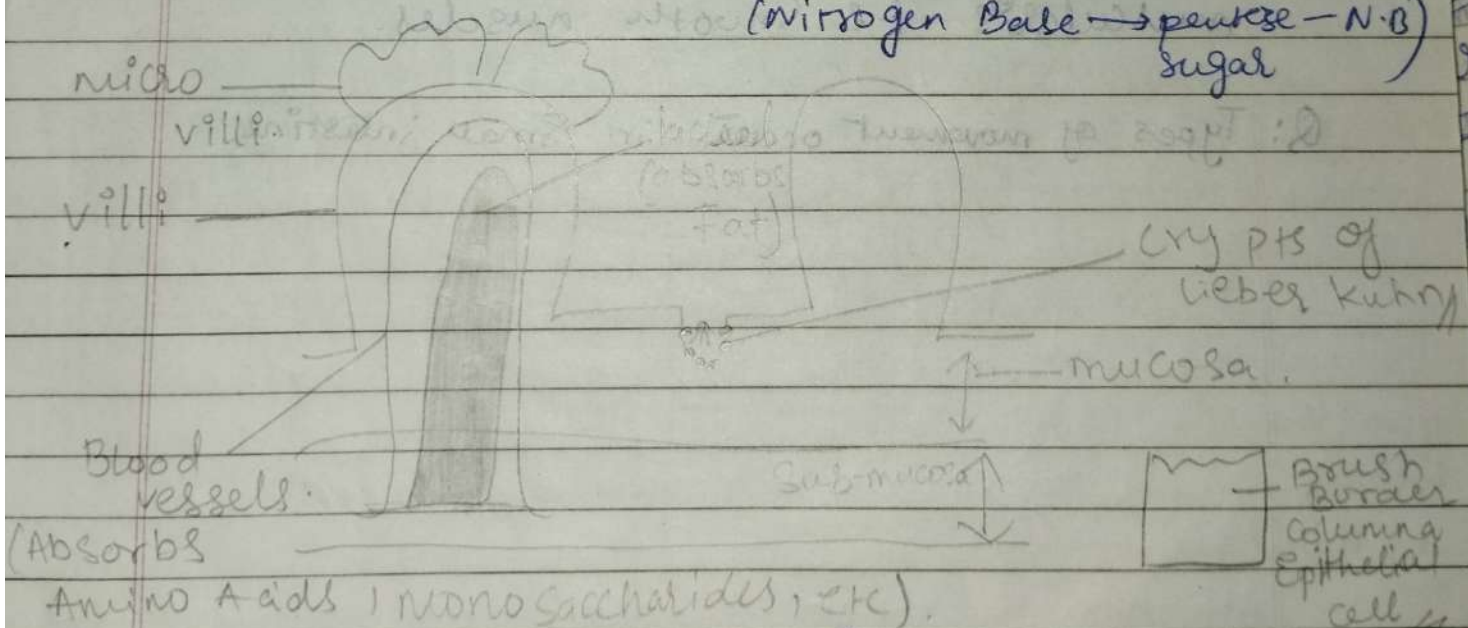
Nucleic acid nuclease → Nucleotides.

DNAse
RNAse.



Nucleosides

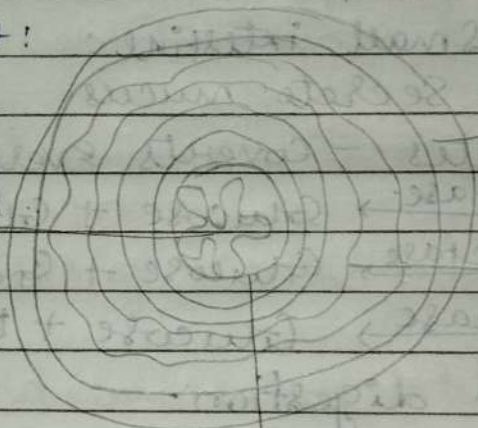
(Nitrogen Base → pentose - N.B) sugar



layers of Small Intestine.

- Large intestine: • Absorption of some water, minerals and some drugs takes place here only.
- Secretion of mucus to lubricate easy passage of feces.

- T.S of Gut:



- outer longitudinal and inner circular make muscularis i.e. smooth muscles.

Q: Types of movement occur in small intestine.